

respect to O . (Remember: $\text{vert}(H)$ is an initial set with respect to O .) Thus, $\text{vert}(F) \subseteq \text{vert}(H)$. Since both H and $G(F)$ are k -regular and connected, $\text{vert}(F) = \text{vert}(H)$ and $G(F) = H$. This completes the proof. $\square$

Remarks 3.13.

0. You could ask: do these parameters h_i^O actually *mean* anything? They do: we will come back to this when we study shellability, in Lecture 8.
1. We do not have a practical way to distinguish between good and bad orientations. The algorithm suggested by the preceding proof is exponential in $|\text{vert}(P)|$, but it can be made to “work” in practice: see Achatz & Kleinschmidt [1]. Is there a really *efficient* way to compute, for example, the number of facets of P from $G(P)$?
2. General polytopes cannot be reconstructed from their graphs — this can be seen, for example, from the existence of neighborly (simplicial) polytopes. However, Joswig [293] has an extension of Kalai’s result and proof to non-simple polytopes.

Perles [435, 437] proved that simplicial d -polytopes are determined by their $\lfloor d/2 \rfloor$ -skeleta; see Kalai [307]. General d -polytopes are determined by their $(d-2)$ -skeleta, and this is best possible even for quasi-simplicial polytopes (all of whose facets are simplicial polytopes); see Grünbaum [252, Ch. 12].

3.5 Balinski's Theorem: The Graph is d -Connected

A very fundamental fact about the graphs of d -polytopes is that they are d -connected, a theorem due to Balinski [37].

Here we use the definition that a simple graph $G(P)$ is *d -connected* if the removal of any $d-1$ or fewer vertices (and all the edges they are incident with) leaves a connected graph.

The theorem is certainly plausible, since it is easy to see (using Lemma 3.6) that every vertex of $G(P)$ has degree at least d . We have adapted the following simple proof from Grünbaum [252]. Two different proofs appear in Brøndsted & Maxwell [134] and in Barnette [50]. Two extensions, which answer the questions of how many components the graph may have if you remove k vertices, or if you remove a k -face, appear in Klee [322] and in Perles & Prabhu [439]. A stronger, “directed” version of Balinski’s theorem is hidden in Holt & Klee [282].

Theorem 3.14 (Balinski’s theorem). [37]

The graph $G(P)$ is d -connected for every d -polytope P .

Proof. Let $P = \text{conv}(V) \subseteq \mathbb{R}^d$, where the vertex set V of P and of the graph $G(P)$ has at least $d+1$ elements. We delete a subset of $d-1$ of them, $S = \{v_1, \dots, v_{d-1}\} \subseteq V$; then we have to show that the graph $G(P) \setminus S$ induced on the remaining vertices is connected.

Let $\mathbf{s} := \frac{1}{d-1} \sum_{i=1}^{d-1} \mathbf{v}_i \in P$ denote the barycenter of the vertex set S . We know by Lemma 2.9(i) that $\mathbf{s}$ is contained in the relative interior of a face F . We consider two cases.

Case 1. If $\mathbf{s}$ is contained in a proper face $F \in L(P) \setminus \{P\}$, then all points $\mathbf{v}_i \in S$ are also contained in this face F — this is the usual computation. Let $\mathbf{c}\mathbf{x} \leq c_0$ be a valid inequality that defines F . Then c_0 is the largest value that $\mathbf{c}\mathbf{x}$ can achieve on P , while the smallest value is some $g_0 < c_0$. In this case, every vertex in $V \setminus S$ either lies in the face $F_0 = \{\mathbf{x} \in P : \mathbf{c}\mathbf{x} = g_0\}$, or it has a neighbor whose $\mathbf{c}\mathbf{x}$ -value is smaller (this follows from Lemma 3.6), and which therefore also lies in $V \setminus S$. Thus every vertex in $V \setminus S$ has a decreasing path, within $V \setminus S$, which connects it to a vertex in F_0 . Finally, the graph of F_0 is connected, by induction on d .

Case 2. If $\mathbf{s}$ is contained in the interior of P , then we choose a linear function $\mathbf{c}\mathbf{x}$ on $\mathbb{R}^d$ such that the hyperplane $\{\mathbf{x} \in \mathbb{R}^d : \mathbf{c}\mathbf{x} = c_0\}$ contains both S and at least one other vertex $\mathbf{v}_0 \in V \setminus S$. This is possible because every set of d points is contained in a hyperplane.

Now let $c_{\max}$ and $c_{\min}$ denote the largest and the smallest value, respectively, that $\mathbf{c}\mathbf{x}$ takes on P , and let $F_{\max}$ and $F_{\min}$ denote the corresponding faces. Then the graphs $G(F_{\max})$ and $G(F_{\min})$ are again connected, by induction. Every vertex $\mathbf{v} \in V \setminus S$ is connected either by a strictly $\mathbf{c}\mathbf{x}$ -increasing path which avoids S to $F_{\max}$ (if it satisfies $\mathbf{c}\mathbf{v} \geq c_0$), or by a strictly decreasing path to $F_{\min}$ (if $\mathbf{c}\mathbf{v} \leq c_0$). Finally, the extra vertex $\mathbf{v}_0$ is connected to both $F_{\max}$ and $F_{\min}$, so the whole graph $G(P) \setminus S$ is connected. $\square$

Notes

The graph of a polytope is treated with care in Grünbaum's book [252, Chapters 11, 13 and 16].

As for linear programming, this is usually described in a much less geometric way, which is better suited for algorithmic treatment. Also, there is of course much more to say than our simplified sketch in Section 3.2. We refer to the books by Dantzig [174], Chvátal [158], Schrijver [484], Grötschel, Lovász & Schrijver [246], Padberg [434], and Borgwardt [125], and to [96, Ch. 10] for various different aspects of the matter. As for “Dantzig's simplex algorithm” [174], let us just mention that it was already developed by Kantorovich in the 1920s, but could not published in any reasonable form for reasons that were equally ideological and stupid [312].

Klee & Kleinschmidt [327] is an inspiring survey on the Hirsch conjecture and its relatives; see also Kleinschmidt [335]. The material in Section 3.3 is derived from the papers by Kalai & Kleitman [305, 309, 306]. In particular, the proof of Theorem 3.10 is from Kalai [306, Sect. 2]. Our discussion of 0/1-polytopes is based on ideas by Naddef [419] and by Kleinschmidt [336]. The observation in Theorem 3.11 that the extreme case here is only achieved for d -cubes seems to be new (although not deep).